WORDS TO PICTURES

SMS spam and ham language - Insight report

Executive summary

Analyzed 5,574 UCI messages: 4,827 ham, 747 spam, 90,381 word-boundary tokens. Largest plotted prevalence gap: call, 45.6% of spam vs 5.6% of ham (+40.0%). These are historical corpus associations, not current detection accuracy. Use cues for human review, audit false positives, and refresh with current labels.

Question, dataset, method

Primary question: what language distinguishes spam from ham and what should filtering teams prioritize? Sub-questions examine terms, length, and context. Source: UCI SMS Spam Collection, DOI 10.24432/C5CC84, CC BY 4.0, accessed 2026-10-27. Empty text/missing labels excluded; duplicates retained; generated IDs. Historical convenience corpus, not representative.

Regex tokens, lowercase, stop-word removal preserving negations, Porter stemming; DTM 5,574 by 6,524; TF-IDF L2-normalized. Original text for KWIC.

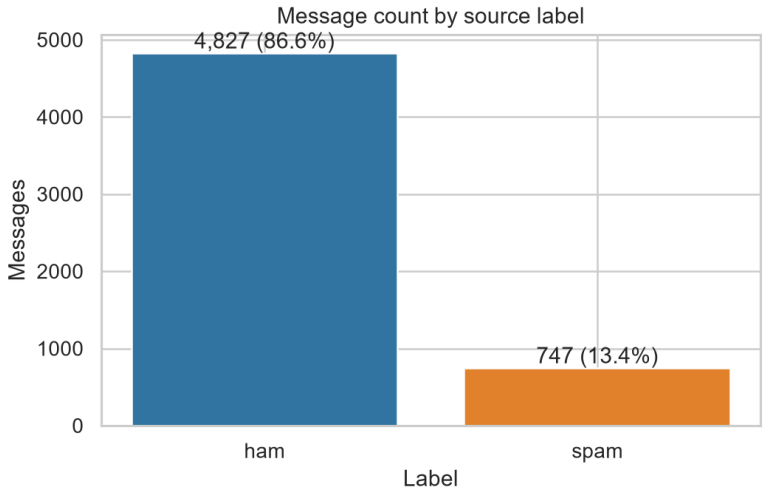


Figure 1. Ham 4,827 (86.6%); spam 747 (13.4%).

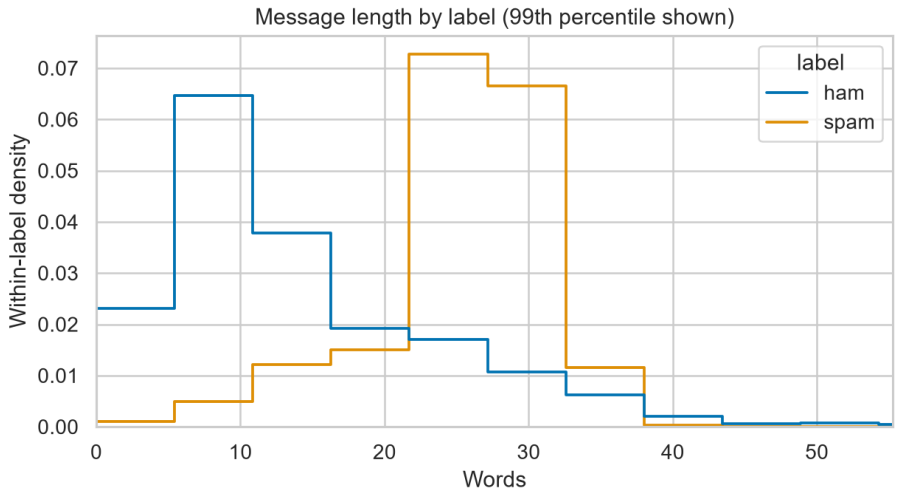


Figure 2. Median ham 11, spam 27 words; display clipped at 99th percentile.

Visual findings

Top positive mean TF-IDF contrast: call (0.0574); most negative: ok (-0.0184). Scores are not probabilities.

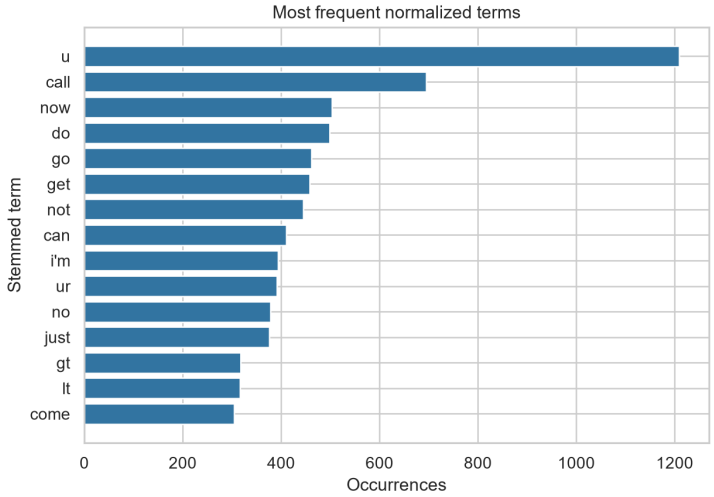


Figure 3. Frequent stemmed terms; raw counts reflect class imbalance.

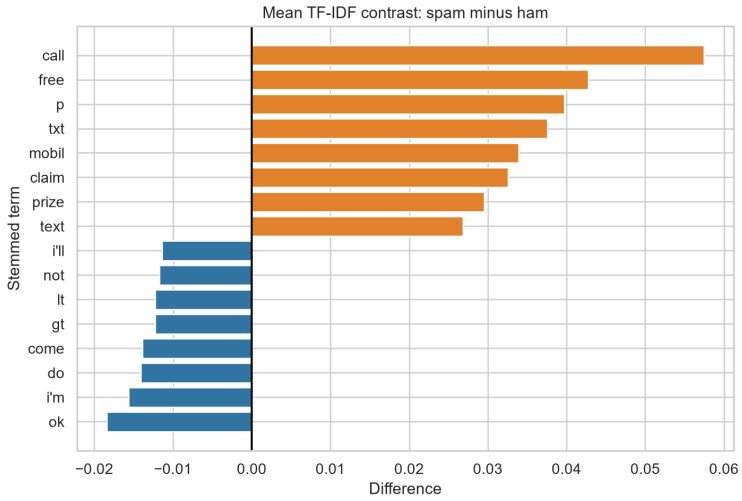


Figure 4. Mean TF-IDF contrast, spam minus ham.

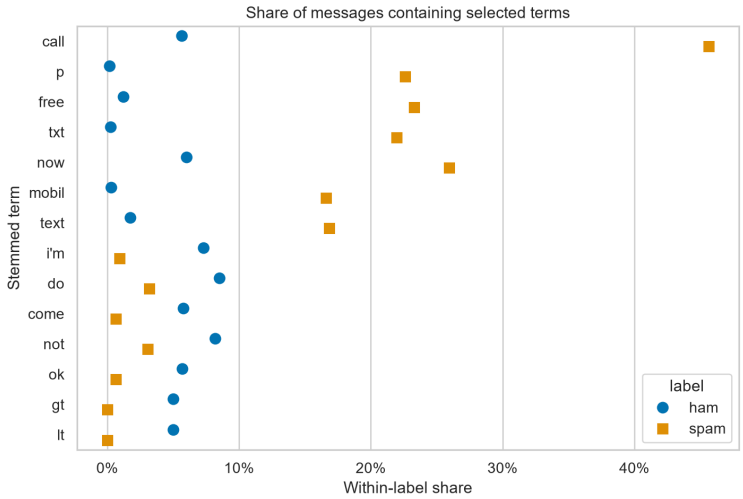


Figure 5. Within-label message share; largest shown gap call: 45.6% vs 5.6%.

Original KWIC contexts for call

SPAM SMS0009: …to receivea £900 prize reward! To claim <<call>> 09061701461. Claim code KL341. Valid 12…

SPAM SMS0010: …st colour mobiles with camera for Free! <<Call>> The Mobile Update Co FREE on 0800298603…

SPAM SMS0043: …4351 - Rodger Burns - MSG = We tried to <<call>> you re your reply to our sms for a free…

SPAM SMS0057: …ear special cinema pass for 2 is yours. <<call>> 09061209465 now! C Suprman V, Matrix3, …

SPAM SMS0066: …u are awarded with a £1500 Bonus Prize, <<call>> 09066364589

SPAM SMS0094: Please <<call>> our customer service representative on …

HAM SMS0008: …e Nurungu Vettam)' has been set as your <<callertune>> for all Callers. Press *9 to copy your …

HAM SMS0046: No <<calls>>..messages..missed calls

HAM SMS0058: Sorry, I'll <<call>> later in meeting.

HAM SMS0075: U can <<call>> me now...

HAM SMS0076: I am waiting machan. <<Call>> me once you free.

HAM SMS0081: Sorry, I'll <<call>> later

Figure 6. Short original contexts for call; first hits are illustrative.

Trust, recommendations, conclusion

Figure 4 is interpretable with a zero-centered scale and faithful to the computation; trust is limited by age, sample, inherited labels, and stemming. Figure 5 has a clear proportion scale, but duplicates may not be independent and terms were selected on the same corpus. Neither validates prediction. Test on current held-out data.

1. Use terms as human-review cues and evaluate precision, recall, false positives. 2. Contextually audit legitimate matches. 3. Repeat on consented current data to monitor drift. Sender fairness cannot be examined here. Results justify evaluation, not automatic rules or causation.

References

Almeida et al. (2011). Contributions to the study of SMS spam filtering. UCI SMS Spam Collection. https://doi.org/10.24432/C5CC84. Scikit-learn: https://scikit-learn.org/. NLTK: https://www.nltk.org/.